



**INTRODUCTORY PHYSICS LABORATORY
DEPARTMENT OF PHYSICS
FACULTY OF SCIENCE AND DATA ANALYTICS
INSTITUT TEKNOLOGI SEPULUH NOPEMBER
SURABAYA**



NAME :
STUDENT ID :
ASSISTANT :
DATE :

EXPERIMENT: MOMENTUM AND COLLISIONS

Objectives:

1. To understand the law of conservation of linear momentum in collisions.
2. To determine the types of collisions based on the coefficient of restitution.

EXPERIMENT SUMMARY

INTRODUCTION

Collision and momentum phenomena are often encountered in everyday life, such as a ball bouncing on the floor, vehicles colliding, or a person throwing and catching a ball. All of these events involve force interactions between two objects that cause changes in motion or direction. According to Giancoli [1], momentum is a vector quantity that describes the magnitude of an object's motion and is directly related to the object's mass and velocity. When two objects collide, the momentum of each object changes, but the total momentum of a closed system remains constant. This principle is known as the law of conservation of momentum, which serves as the foundation for analyzing various collision events [1].

The relationship between momentum and collision is closely related to the concept of impulse, which is the product of force and the time interval during which the force acts on an object. According to the impulse-momentum theorem, the impulse applied to an object is equal to the change in momentum experienced by that object. The change in momentum during a collision is influenced by the magnitude of the force and the duration of contact between the objects. The greater the force or the longer the force acts, the greater the impulse produced, resulting in a larger change in momentum. This impulse concept provides a quantitative basis for explaining the momentum changes that occur during the collision process [4].

Momentum is one of the fundamental concepts in physics that describes the tendency of an object to maintain its motion. If an object of mass m moves with a constant velocity v , then the momentum of the object is expressed by the following equation.

$$P = mv \tag{1}$$

When two objects collide, the total momentum of the system before and after the collision remains constant as long as no external forces act on the system. This principle is known as the law of conservation of linear momentum, which is expressed by the following equation.

$$P_{\text{sebelum tumbukan}} = P_{\text{sesudah tumbukan}} \tag{2}$$

$$m_1v_1 + m_2v_1 = m_1v'_1 + m_2v'_2 \tag{3}$$

where m_1 is the mass of object 1, m_2 is the mass of object 2, v_1 is the velocity of object 1 before collision, v_2 is the velocity of object 2 before collision, v'_1 is the velocity of object 1 after collision, and v'_2 is the velocity of object 2 after collision [2]-[3].

In general, collisions are classified into three types: perfectly elastic collisions, partially elastic collisions, and perfectly inelastic collisions. These three types of collisions can be distinguished based on the coefficient of restitution, which is expressed by the following equation.

$$e = \left| \frac{\Delta v'}{\Delta v} \right| \tag{4}$$

The classification of collision types based on the coefficient of restitution is presented in Table 1.

Table 1. Types of collisions and their coefficient of restitution (e) values.

Type of Collision	Coefficient of Restitution (e)
Perfectly inelastic	$e = 0$
Partially elastic	$0 < e < 1$
Perfectly elastic	$e = 1$

The coefficient of restitution value indicates the characteristic of each type of collision based on the ratio of relative velocities before and after the collision. The larger the coefficient of restitution, the less kinetic energy is lost during the collision process [5].

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 2. Tools and materials for the momentum and collisions experiment.

Tools	Quantity	Function
Momentum and collisions	1 set	Used to conduct the momentum and collisionsexperiment.
Loads	1 set	Used to provide mass variations on the colliding objects.
Velocity sensors	2 pieces	Used to measure the velocity of objects before and after collision.
CASSY device	1 piece	Used to record data from the sensors to the Laptop.
Laptop	1 piece	Used to record and process data.
Dimmer	1 piece	Used to regulate the airflow speed from the vacuum cleaner.
Vacuum cleaner	1 piece	Used to generate airflow to reduce friction on the track.
Digital scale	1 piece	Used to measure the mass of the loads.

B. Procedure

The steps for this experiment are as follows.

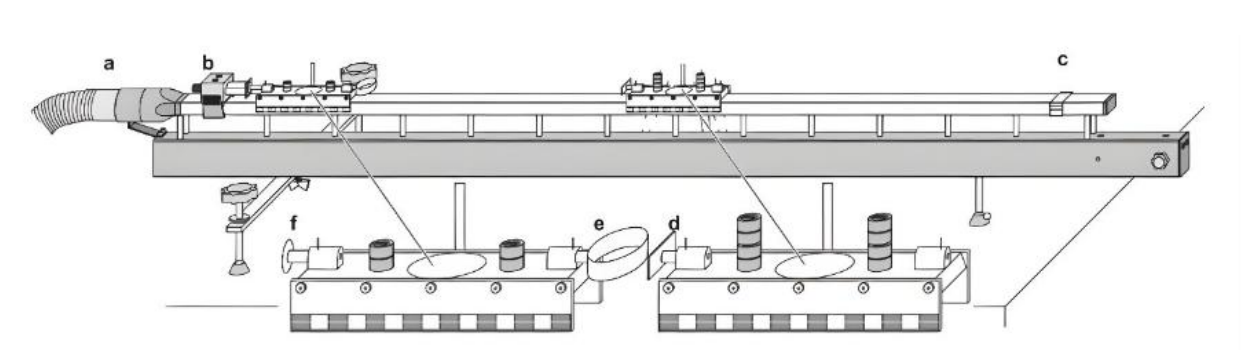


Figure 1. Experimental setup of the momentum and collisions.

1. Set up the experimental setup as shown in Figure 1.
2. Connect the dimmer to the PLN power source.
3. Connect the vacuum cleaner to the dimmer.
4. Place the velocity sensors at a predetermined distance.
5. Connect the velocity sensors to the CASSY device and connect the CASSY device to the PLN power source.
6. Connect the CASSY device to a laptop that has CASSY Lab 2 software installed.
7. Open the CASSY Lab 2 software and ask the assistant to open the required page.
8. Weigh the masses of the loads on the carts using a digital balance, then place the loads on cart 1 and cart 2.
9. Enter the load mass data into the CASSY Lab 2 software.
10. Position carts 1 and 2 at a predetermined distance.
11. Turn on the vacuum cleaner and dimmer, adjust the dimmer knob to regulate the airflow speed so that the carts do not touch the track.
12. Push carts 1 and 2 simultaneously, then record the velocity data displayed on the CASSY Lab 2 software.
13. Repeat step 12 ten times as repetitions with the same load mass on the carts.
14. Repeat steps 8 – 13 with different load mass variations on the carts.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement result obtained in the laboratory:

Table 3. Measurement results for velocity at a load mass variation of 1:1.

Repetition	m_1 (kg)	m_2 (kg)	v_1 (m/s)	v_2 (m/s)	v'_1 (m/s)	v'_2 (m/s)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Table 4. Measurement results for velocity at a load mass variation of 1:2.

Repetition	m_1 (kg)	m_2 (kg)	v_1 (m/s)	v_2 (m/s)	v'_1 (m/s)	v'_2 (m/s)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Table 5. Measurement results for velocity at a load mass variation of 1:3.

Repetition	m_1 (kg)	m_2 (kg)	v_1 (m/s)	v_2 (m/s)	v'_1 (m/s)	v'_2 (m/s)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Approved by:

Laboratory Assistant

B. Calculation Analysis

Calculate the momentum before and after the collision, the coefficient of restitution, and the absolute ratio of momentum after to before the collision along with their measurement uncertainties for all mass variations using Equations (1), (3), and (4)! Determine the type of collision based on the coefficient of restitution value obtained!

The following is a sample calculation based on the measurement data.

Sample calculation:

The following is the table of calculation results obtained:

Table 6. Calculation results for momentum and coefficient of restitution at a load mass variation of 1:1.

Repetition	p_o (kg.m/s)	p' (kg.m/s)	Absolute Ratio $\left(\frac{p'}{p_o}\right)$	e	Type of Collision
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
Average					
Standard Deviation					
Uncertainty					

Table 7. Calculation results for momentum and coefficient of restitution at a load mass variation of 1:2.

Repetition	p_o (kg.m/s)	p' (kg.m/s)	Absolute Ratio $\left(\frac{p'}{p_o}\right)$	e	Type of Collision
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
Average					
Standard Deviation					
Uncertainty					

Table 8. Calculation results for momentum and coefficient of restitution at a load mass variation of 1:3.

Repetition	p_o (kg.m/s)	p' (kg.m/s)	Absolute Ratio $\left(\frac{p'}{p_o}\right)$	e	Type of Collision
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
Average					
Standard Deviation					
Uncertainty					

D. Discussion

CONCLUSION

From this experiment, it can be concluded that:

1.

2.

REFERENCES

- [1] Giancoli, D.C. (2014). *Physics: Principles with Applications* (7th ed.). Upper Saddle River, NJ: Pearson Education.
- [2] Halliday, D., Resnick, R. and Krane, K.S. (2010). *Fundamentals of Physics* (8th ed.). Hoboken, NJ: John Wiley & Sons.
- [3] Halliday, D., Resnick, R. and Walker, J. (2010). *Fundamentals of Physics Extended* (9th ed.). Hoboken, NJ: John Wiley & Sons.
- [4] Serway, R.A. and Jewett, J.W. (2012). *Physics for Scientists and Engineers with Modern Physics* (9th ed.). Mason, OH: Brooks/Cole, Cengage Learning.
- [5] Young, H. D., & Freedman, R. A. (2016). *University Physics with Modern Physics* (14th ed.). Pearson.

PRELIMINARY ASSIGNMENT

Answer the following questions before conducting the experiment.

1. What is meant by the law of conservation of momentum? Write its equation!
2. Explain the types of collisions based on the coefficient of restitution (e)! Include their characteristics and examples!
3. In the momentum and collision experiment, why is it important to use an air track or a smooth surface? Explain its relation to the law of conservation of momentum!
4. Two objects with masses $m_1 > m_2$ move with the velocity of the first object being twice that of the second object ($v_1 = 2v_2$) and undergo a perfectly elastic collision. Prove mathematically that the velocity of the second object after collision (v'_2) is greater than the velocity of the first object after collision (v'_1)!
5. If the collision in question number 4 is a perfectly inelastic collision, prove mathematically that both objects will move together with velocity $\left(v' = \frac{2m_1v + m_2v}{m_1 + m_2}\right)$ and explain its physical meaning!